

6	(a)	Magnetic field lines are directed <u>out of the paper</u> . [A1]
	(b)	Magnetic force provides for centripetal force $Bqv = m\omega^2 r \quad [\text{M1}]$ where ω is the angular velocity of the proton. $m\omega^2 r = Bq\omega r$ $\omega = \frac{Bq}{m}$ $\therefore \omega = \frac{2\pi}{T} \quad [\text{M1}]$ where T is the period of the motion. $T = \frac{2\pi m}{Bq} \quad [\text{M1}]$ $t_D = \frac{1}{2}T = \frac{\pi m}{Bq} \quad [\text{A0}]$



		Alternative working: Magnetic force provides for centripetal force $Bqv = m \frac{v^2}{r} \quad [\text{M1}]$ $r = \frac{mv}{Bq} \quad [\text{M1}]$ $t_D = \frac{\pi r}{v} \quad [\text{M1}]$ $= \frac{\pi m}{Bq} \quad [\text{A0}]$	
	(c)	The magnetic field does not change the speed of the proton. The electric field causes the speed to increase.	[B1] [B1]
	(d)	$t_D \text{ or } \frac{\pi m}{Bq}$	[A1]
	(e)	Since the charge of the new particle is negative: This particle will make an anticlockwise direction motion within the dees. [B1] Since the mass-to-charge ratio (m/q) of the new particle is the same: This particle will make the same radius within the dees [B1] Since m/q of this new particle is the same: Based on the equation of (b) the t_D value will remain same since B remains the same. [B1]	